# Rethinking Visual Features in Graph Based Multimodal Emotion Recognition in Conversations (MERC): From Low Level Pixels to Semantic Representations

Anonymous ECCV 2026 Submission

Paper ID #14677

**Abstract.** Multimodal Emotion Recognition in Conversations (MERC) aims to identify emotions in each utterance using visual, audio, and linguistic cues, with applications in affective computing and human-computer interaction. Among the various approaches proposed to solve this problem, graph based approaches have emerged as some of the most promising ones but they often underutilize the visual modality due to noisy pixel level features. Traditional visual encoders focus on geometric and texture patterns, which often fail to capture facial expressions and behavioral cues, resulting in weak cross-modal alignment. To address this issue, we propose a Vision Language Model (VLM) based approach that encodes facial expressions and behavioral cues from the visual modality into semantically meaningful representations. Each utterance's video is translated into a descriptive textual representation using structured VLM prompts capturing expressions, gestures, body language and scenario understanding. These semantic visual features replace traditional low level visual inputs in graph based MERC frameworks, reducing noise and allowing the model to focus on meaningful visual information. This work presents VLM based semantic visual feature extraction as a viable replacement for conventional visual feature extraction pipelines in graph based MERC frameworks.

**Keywords:** Multimodal Emotion Recognition in Conversations · Vision Language Models · Graph Neural Networks

## 1 Introduction

The proper interpretation and communication of emotions are crucial, as they significantly influence interactions, conversations, and decision-making processes. In an AI-driven future, it is essential that emerging systems can recognize emotions at or above human ability. Understanding emotions in conversations remains a fundamental challenge in human–computer interaction and affective computing [5].

Multimodal Emotion Recognition in Conversations (MERC) aims to identify emotions in each utterance by leveraging textual, acoustic, and visual signals

[28,36]. Unlike single-utterance emotion recognition, MERC considers contextual dependencies across speakers and time, making it a more realistic and complex problem [10,24]. Approaches range from early fusion techniques and transformer-based models to more structured, context-aware graph based architectures [10, 14, 32, 34, 35].

Graph based architectures, inspired by Graph Convolutional Networks (GCNs) [17], are particularly effective for MERC [10, 14]. By modeling conversations as graphs with utterances as nodes and relational dependencies as edges, these models capture both intra and inter-speaker interactions, achieving competitive performance [14].

Despite these advances, visual features remain underutilized in many graph based MERC frameworks [14, 19]. Traditional visual encoders such as CNNs [13] or Vision Transformers (ViTs) [7] extract low level spatial or patch based representations, which often fail to capture subtle behavioral cues and body language necessary for emotion understanding [21]. Aggregating frame level embeddings into utterance level features further collapses temporal dynamics, averaging out expressive shifts within an utterance [34], weakening cross-modal alignment compared to textual or acoustic modalities [6, 31].

To address this limitation, we propose a Vision Language Model (VLM)-based approach that encodes visual content as semantically meaningful descriptions. Using VidEmo-3B (based on Qwen2.5-VL-3B [1]), each utterance-level video is translated into a structured textual narrative capturing facial expressions, head movements, and body language. These high-level semantic features preserve intra-utterance dynamics and align better with linguistic and acoustic modalities [29]. Replacing conventional low level visual embeddings with these features in graph based MERC architectures reduces noise, improves cross-modal consistency, and encourages the model to leverage meaningful visual cues.

In summary, this work makes the following contributions:

1. We identify and analyze the underutilization of the visual modality in graph based MERC due to low level, noisy feature representations.
2. We propose a VLM-based semantic visual feature extraction approach that converts raw video into structured textual descriptions capturing emotion-relevant cues.
3. We integrate these semantic visual features into graph based MERC architectures, promoting stronger cross-modal alignment.

By bridging the gap between visual perception and semantic understanding, our approach offers a new direction for enhancing the role of visual information in multimodal conversational emotion recognition.

## 2 Related Works

### 2.1 Emotion Recognition in Conversations and Multimodal Extensions

Emotion Recognition in Conversations (ERC) aims to identify the emotional state of each utterance by leveraging dialogue context and speaker interac-

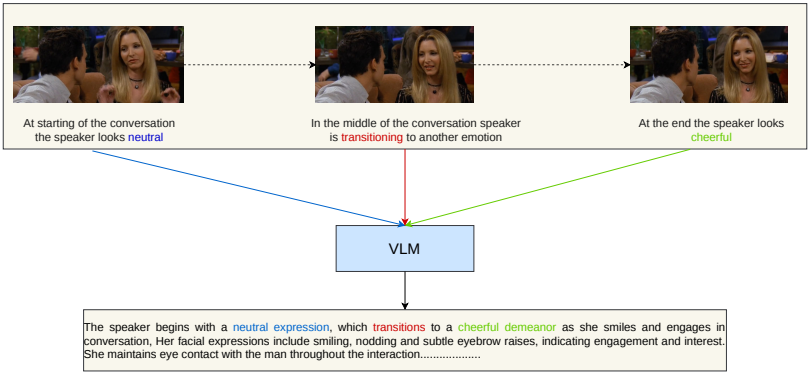


**Fig. 1:** Overview of the proposed semantic visual preprocessing pipeline.

tions [32]. Early sequence models using RNNs or GRUs capture short term dependencies and speaker states [11, 24], but often struggle with long range interactions.

Multimodal ERC (MERC) incorporates text, acoustic, and visual cues to better reflect human emotion expression [28, 36]. Text provides lexical and semantic content, acoustic features convey speech patterns [31], and visual data encodes facial expressions and gestures [21]. MERC introduces additional challenges, including modality alignment, noise handling, and effective cross-modal fusion [32].

### 2.2 Approaches to MERC

To address these challenges, several modeling strategies have been proposed.

**Sequence and Context Based Methods.** Sequence models capture temporal dependencies and speaker states in conversations. DialogueRNN [24] uses recurrent structures to leverage dialog history for emotion prediction but has limited ability to model complex speaker or cross-modal interactions.

**Fusion Based Methods.** Fusion approaches integrate multimodal information using different strategies. Early and late fusion methods combine modality features either at the input level or at the decision stage and are widely used baselines in multimodal sentiment and emotion analysis [25–27, 37]. More advanced approaches employ attention mechanisms to model interactions between modalities. Multimodal Transformer architectures use cross modal attention to align textual, acoustic, and visual representations [34], while MISA learns modality invariant and modality specific representations for improved multimodal integration [12]. Additionally, models such as DialogueRNN and related methods

incorporate mechanisms to track speaker states and emotion transitions within conversations [11, 24].

Although fusion methods improve multimodal interaction, they often do not explicitly represent conversational structure or relational dependencies between speakers.

### 2.3 Graph Based MERC Approaches

Graph based methods are powerful for MERC as they encode conversational structure and relational dependencies [14, 19]. Utterances are nodes and edges represent speaker interactions, temporal continuity, and cross-modal links [17]. DialogueGCN [10] introduced graph reasoning for ERC, modeling intra- and inter-speaker relations and outperforming sequential models. Extensions like MMGCN [14] and hypergraph models capture multimodal interactions, while **Joyful** [18] combines graph fusion with contrastive learning to improve multimodal representations and address issues like oversmoothing.

**GraphSmile** [19] represents a state-of-the-art graph-centric MERC architecture introducing two key components: *Graph Structure Fusion (GSF)* [19], which constructs bipartite dialogue graphs [14] between every pair of modalities and alternates information propagation across these graphs to prevent any single structural path from dominating; and *Sentiment Dynamics Perception (SDP)* [19], an auxiliary task that explicitly predicts whether sentiment shifts occur between consecutive utterances to guide the model toward emotional transitions. Despite its strong performance, GraphSmile, like most existing graph based MERC methods depends on conventional visual encoders (*e.g.*, CNN backbones [13] or handcrafted facial features [2]) that produce static, low level visual embeddings. These embeddings are typically mean-pooled over frames, collapsing intra utterance temporal dynamics and limiting the visual modality's contribution to emotional reasoning. Graph based approaches excel at structurally modeling dialogues and multimodal interactions, but their effectiveness is often constrained by the quality of unimodal representations feeding into the graph.

### 2.4 Vision Language and LLM Based Approaches for Emotion Recognition

To overcome the limitations of low-level visual features, recent research has explored Vision Language Models (VLMs) and multimodal Large Language Models (LLMs) for emotion recognition. Pretrained VLMs [29] generate semantically meaningful descriptions from visual inputs, capturing expressions, behaviors, and dynamics that CNN [13] or ViT [7] features often miss. These textual representations are inherently aligned with language modalities [6], facilitating cross-modal fusion and multimodal reasoning.

VLMs such as VidEmo-3B [38] produce temporally grounded visual narratives, describing affective cues across an utterance. Task-specific adaptations like Emotion-Qwen [15] integrate emotion reasoning with general vision-language backbones, combining specialized emotion experts with general VL features.

With appropriate prompting or fine-tuning, these models capture subtle emotional expressions and behaviors that traditional encoders overlook.

Multimodal LLMs extend this concept further. Frameworks like BeMERC [9] leverage behavior-aware LLMs to integrate visual and linguistic context for conversational emotion prediction. Prompting general VL models such as CLIP [29] or Qwen2.5-VL [1] can enable zero-shot emotion inference from video or images without retraining.

Overall, VLM/LLM approaches address two core MERC challenges: they reduce visual modality underutilization by producing high-level semantic descriptions [21], and improve cross-modal alignment by encoding visual and linguistic information in shared semantic spaces [29]. Integrating these semantic visual representations into graph-based MERC frameworks represents a promising direction for enhanced multimodal emotion recognition.

### 2.5 Positioning of Our Work

Our work builds upon the strengths of graph based MERC frameworks , particularly GraphSmile [19], while addressing the underutilization of visual information. Instead of using static, low level visual embeddings [2, 13], we introduce a VLM based semantic visual preprocessing stage [1, 38] that converts raw video segments into temporally structured textual narratives capturing facial expressions, gaze shifts, and body language evolution within an utterance.

These semantic visual descriptions are encoded into dense vectors [6] and integrated into the Graph Structure Fusion (GSF) pipeline [19]. This design preserves intra utterance temporal dynamics, improves cross modal alignment, reduces reliance on noisy visual appearance features, and enhances the model's ability to leverage visual cues in relational reasoning across conversation turns. By combining high level semantic visual reasoning from VLMs with structured graph propagation, our approach advances MERC beyond existing graph based and VLM based methods.

## 3 Methodology

The proposed framework builds upon the GraphSmile architecture [19] by introducing a novel approach that converts raw visual inputs into meaningful semantic representations before multimodal fusion.

While prior graph based Multimodal Emotion Recognition in Conversation (MERC) systems relied on conventional visual encoders (such as DenseNet for extracting abstract spatial features [16] and OpenFace for tracking deterministic facial mechanics like Action Units [2]), our methodology moves beyond these restricted appearance descriptors. By leveraging a temporally aware Vision Language Model (VLM) [1,29], we encode the rich, semantically grounded behavioral signals present in human speech.

The overall architecture comprises three main stages:

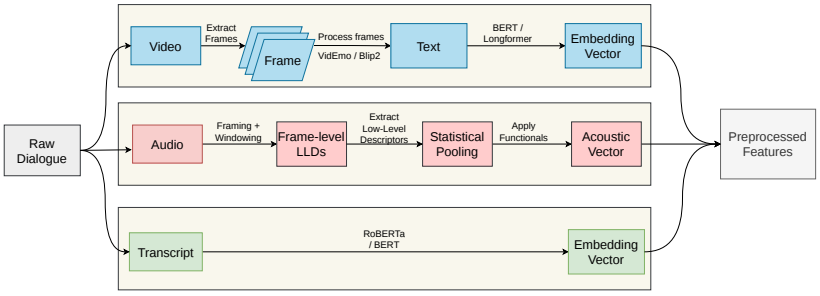


**Fig. 2:** The multimodal preprocessing pipeline for extracting utterance-level textual, acoustic, and visual features.

1. The independent extraction of utterance level features across textual, acoustic, and visual modalities.
2. The construction of multimodal dialogue graphs for alternating propagation via the Graph Structure Fusion (GSF) module [19].
3. Multitask classification incorporating a Sentiment Dynamics Perception (SDP) module [19].

### 3.1 Utterance Level Feature Extraction

Each modality is processed independently at the utterance level before being integrated into the multimodal graph. In contrast to the visual channel, the textual and acoustic modality remains structurally unchanged relative to the GraphSmile baseline [19], preserving a controlled point of comparison.

**Textual Feature Extraction.** For the textual modality, utterance transcripts are encoded uniformly across both datasets using a pretrained transformer language model, specifically RoBERTa [22]. The `<s>` (classification) token representation of each transcript forms the textual feature vector, $X^t$, capturing lexical and syntactic sentiment signals.

**Acoustic Feature Extraction.** The acoustic features, $X^a$, capture paralinguistic speech characteristics including fundamental frequency (F0), energy, spectral properties, and temporal statistical functionals that reflect prosody and pitch dynamics. For both the IEMOCAP (6-way) [4] and MELD [28] datasets, these utterance level features are extracted using the openSMILE toolkit with the IS10 configuration [8].

**Semantic Visual Feature Extraction** Traditional visual encoders often fail to capture conversational cues, as they focus on context-blind spatial or facial features, creating a semantic gap between raw movements and emotional states. We address this via a three-stage semantic preprocessing pipeline.

**Stage 1: Dataset-Specific Video Preprocessing.** For IEMOCAP [4], continuous multi-speaker sessions are segmented into utterance-level clips and spatially cropped to isolate the active speaker. MELD [28] already provides pre-segmented clips, so speaker focus is achieved via zero-shot prompting, instructing the VLM to attend only to the active speaker.

**Stage 2: VLM Temporal Analysis.** Utterance clips are processed by VidEmo-3B [38] (Qwen2.5-VL-3B backbone [1]) with structured prompts to focus on the active speaker. Each clip is divided into beginning, middle, and end segments, and analyzed for:

- **Facial Expressions:** Brow, lips, eyes, jaw, and cheek movements.
- **Head & Gaze:** Tilts, nods, shakes, and eye direction.
- **Body Language:** Posture, gestures, and tension/relaxation.
- **Instantaneous Emotion:** Likely emotional state from observed cues.

A one-sentence summary captures the overall emotional arc (e.g., *neutral* → *frustration* → *brief smile*). This produces a dense, temporally aware textual description, avoiding noise from frame averaging.

For MELD, BLIP-2 [20] is used in a frame-level approach: three frames (start, middle, end) are processed individually with emotion-focused prompts, and their outputs concatenated into a single utterance-level description, reducing multi-party noise.

**Stage 3: Text Encoding of VLM Output.** The VLM text is converted to a dense vector via pretrained transformers. BERT [6] extracts the `[CLS]` token from truncated inputs, while Longformer [3] processes the full description. Both yield 768-dimensional embeddings representing the visual modality, $X^v$, in the same semantic space as transcripts, improving cross-modal alignment within the GSF module [19].

### 3.2 Graph Structure Fusion (GSF) Module

The Graph Structure Fusion (GSF) module processes dialogue graphs layer by layer, alternating between sequential edges for temporal flow and speaker edges for personal consistency. This back-and-forth approach prevents any single relationship from dominating and effectively separates a speaker's internal emotional shifts from the conversation's external influences.

**Multimodal Dialogue Graphs** To understand how utterances relate to each other across different sensory inputs, we build three bipartite dialogue graphs [14, 19]. Each graph connects two specific modalities. For example, the textual visual graph is defined as $\mathcal{G}^{tv} = (\mathcal{N}^t, \mathcal{N}^v, \mathcal{E}^{t\leftarrow v}, \mathcal{E}^{v\leftarrow t})$. Here, $\mathcal{N}^t$ and $\mathcal{N}^v$ stand for the sets of textual and visual nodes. The edge sets $\mathcal{E}^{t\leftarrow v}$ and $\mathcal{E}^{v\leftarrow t}$ represent the connections directed from visual to textual nodes, and vice versa.

- **Nodes:** Every utterance $u_i$ is represented by a textual node $n_i^t$ and a visual node $n_i^v$.

- **Edges:** We only draw edges between nodes of opposite modalities. To keep the graph from becoming too dense, we use a sliding window approach. A textual node $n_i^t$ connects to visual nodes from the past $P$ steps (i.e., $n_{i-P}^v, \ldots, n_{i-1}^v$) and the future $F$ steps (i.e., $n_{i+1}^v, \ldots, n_{i+F}^v$).
- **Edge Weights:** These connections are directed to show the specific flow of information. We assign trainable weights to these edges, which start at a value of 1 and are fine tuned as the model trains.
- **Adjacency Matrix:** All of these node and edge relationships are stored in an adjacency matrix. This matrix is the core structural input for the graph neural network.

**Alternate Graph Propagation** Graph Neural Networks (GNNs) [17, 30] are highly effective at uncovering both cross modal and intra modal relationships. In our design, each dialogue graph passes information through multiple simplified convolutional layers. It alternates between extracting inter modal and intra modal features layer by layer, calculated as:

$$X^{(l)} = AX^{(l-1)}\Theta^{(l)}, \quad l = 1, 2, \ldots, L$$

Here, $A$ is the dialogue graph's adjacency matrix, $X^{(l)}$ is the output feature at layer $l$, and $\Theta^{(l)}$ is the trainable weight matrix for that layer. For the textual-visual graph, the starting input $X^{tv,(0)}$ simply combines the raw textual features $[x_1^t, \ldots, x_M^t]$ and visual features $[x_1^v, \ldots, x_M^v]$.

Next, we use residual connections [13] to sum the feature representations from every layer:

$$X = X^{(0)} + \mathrm{FC}^{(1)}(X^{(1)}) + \cdots + \mathrm{FC}^{(L)}(X^{(L)})$$

The fully connected layer $\mathrm{FC}(\cdot)$ includes a linear transformation, a LeakyReLU [23] activation, and dropout [33]:

$$\mathrm{FC}(X) = \mathrm{Dropout}(\mathrm{LeakyReLU}(X\Theta_f + b_f))$$

This setup serves two purposes: it lets the network dynamically choose which layer's information is most useful, and it boosts the overall expressive power of the GSF module.

**Multimodal Integration** After processing the textual-visual ($\mathcal{G}^{tv}$), textual-acoustic ($\mathcal{G}^{ta}$), and visual-acoustic ($\mathcal{G}^{va}$) graphs through their own separate GSF modules, we get three distinct outputs: $X^{tv}$, $X^{ta}$, and $X^{va}$. We add these together to create a final, unified feature representation:

$$\begin{aligned} H = \sigma\big((X^{t\leftarrow v})\Theta_h\big) + \sigma\big((X^{v\leftarrow t})\Theta_h\big) \\ + \sigma\big((X^{t\leftarrow a})\Theta_h\big) + \sigma\big((X^{a\leftarrow t})\Theta_h\big) \\ + \sigma((X^{v\leftarrow a})\Theta_h) + \sigma((X^{a\leftarrow v})\Theta_h) \end{aligned} \tag{1}$$

In this equation, $\sigma$ is the activation function (like LeakyReLU) and $\Theta_h$ is a shared learning parameter. This combined matrix $H$ effectively captures all the complex, multi sensory emotional signals in the conversation.

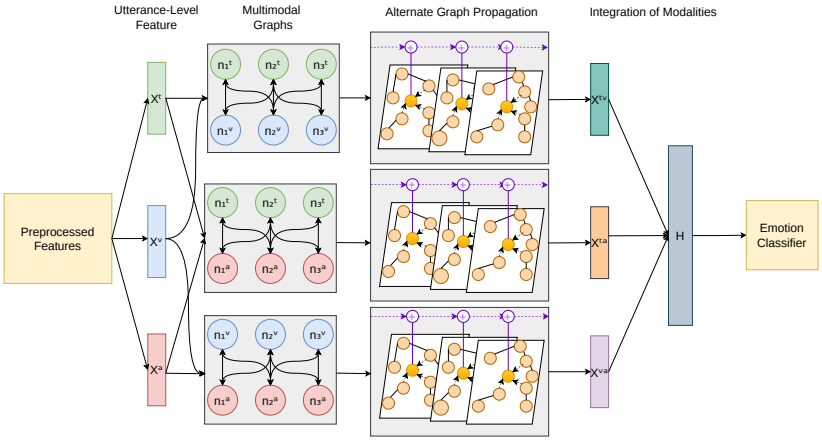


**Fig. 3:** Multimodal dialogue graph structure used in GraphSmile [19], in which utterance nodes from different modality pairs are connected via bipartite graphs and propagated through the Graph Structure Fusion (GSF) module. Adapted from [19].

### 3.3 Emotion Classification

After the graph propagation and integration phase, the resulting feature matrix $H$ is passed into a multitask classification head. For the primary task of emotion classification, $H$ goes through a linear classifier and a softmax function to predict the exact emotional category. This is formulated as:

$$\hat{Y} = \text{softmax}(H\Theta_y + b_y), \qquad \hat{e}_i = \text{argmax}(\hat{y}_i[\tau])$$

Here, $\Theta_y$ and $b_y$ are the learned weights and biases. The vector $\hat{y}_i$ holds the emotion probabilities for utterance $u_i$, and $\hat{e}_i$ is the final predicted emotion label. We optimize the network using categorical cross-entropy loss:

$$\mathcal{L}_e = -\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{C_e} y_{i,j}\log\hat{y}_{i,j}$$

where $M$ is the total number of utterances, $C_e$ is the number of emotion classes, $y_{i,j}$ is the true label, and $\hat{y}_{i,j}$ is our model's prediction.

### 3.4 Sentiment Dynamics Perception (SDP) Module

To capture inter-utterance sentiment changes overlooked by static context models, we adopt the Sentiment Dynamics Perception (SDP) module [19] as an auxiliary task. It casts conversation dynamics as pairwise shift prediction, jointly optimizing a shift loss with the main emotion classification loss to improve class separation and align naturally with the temporal descriptions from the VLM stage.

**Shift Label** We define a sentiment shift label $o_{(i,j)}$ for any pair of utterances $u_i$ and $u_j$. If the sentiment changes between the two utterances, the label is 1; if the sentiment remains the same, the label is 0:

$$o_{(i,j)} = \begin{cases} 1, & \text{if } s_i \neq s_j, \\ 0, & \text{if } s_i = s_j. \end{cases}$$

where $s_i$ denotes the ground truth sentiment state of utterance $u_i$.

**Shift Classification** We construct pairwise shift features $T$ from the fused representations and pass them through a softmax classifier to predict the shift probability $\hat{Z}$:

$$\hat{Z} = \text{softmax}(T\Theta_z + b_z)$$

where $\Theta_z$ and $b_z$ are learnable parameters. The module is optimized using a standard cross-entropy loss function, denoted as $\mathcal{L}_o$.

### 3.5 Sentiment Classification

To extend our model to the Multimodal Sentiment Analysis in Conversation (MSAC) task, we add a sentiment classifier. The fused feature matrix $H$ is processed through a linear layer and a softmax activation:

$$\hat{W} = \text{softmax}(H\Theta_w + b_w)$$

Here, $\Theta_w$ and $b_w$ are learnable parameters, and $\hat{W}$ contains the predicted sentiment probabilities. This task is supervised by a sentiment classification cross-entropy loss, denoted as $\mathcal{L}_s$.

### 3.6 Total Training Objective

To simultaneously extract both emotional and sentimental cues, the primary emotion recognition task is jointly learned alongside the sentiment and shift classification tasks. The overall network parameters are optimized using a combined loss function:

$$\mathcal{L}_{total} = \mathcal{L}_e + \lambda_s \mathcal{L}_s + \lambda_o \mathcal{L}_o + \beta ||\Theta_{total}||$$

where $\mathcal{L}_e$ is the primary emotion classification loss, $\lambda_s$ and $\lambda_o$ are trade-off hyperparameters balancing the auxiliary tasks, $\beta$ is the $L_2$ regularization weight, and $\Theta_{total}$ represents all trainable parameters in the model.

## 4 Experimental Details

### 4.1 Datasets Overview

**IEMOCAP** [4]**.** The dataset consists of continuous, dyadic conversational sessions recorded with a dual-camera setup, presenting both actors side-by-side in a

single unsegmented video frame. The accompanying audio is recorded in stereo, capturing each speaker on a dedicated independent microphone channel.

**MELD** [28]**.** This dataset is constructed from episodes of the television series *Friends*, representing a multi-party conversational environment. The video data is inherently provided as pre-segmented, utterance-level clips.

**Table 1:** Dataset statistics for IEMOCAP and MELD.

| Dataset | Dialogues | Utterances | Classes | Train / Val / Test |
|---|---|---|---|---|
| IEMOCAP | 151 | 7,433 | 6 | 5,810 / —/ 1,623 |
| MELD | 1,432 | 13,708 | 7 | 9,989 / 1,108 / 2,611 |

### 4.2 Utterance-Level Video Preparation

**IEMOCAP.**

- Segment each conversation into utterance level clips using timestamps.
- Crop frames to isolate the active speaker.
- Retain only the speaker's audio channel from the stereo stream.
- Output speaker focused utterance level video clips.

**MELD.**

- Pre-segmented utterance level clips; no cropping needed.
- Each clip corresponds to a single speaker utterance.
- Speaker focus is handled during prompting.

### 4.3 Semantic Visual Feature Extraction

VidEmo-3B [38] (Qwen2.5-VL backbone [1]) generates utterance-level descriptions for both datasets. IEMOCAP is sampled at 1 FPS; MELD uses 1 FPS up to 15 s, capped at 16 frames. Prompts request a temporal progression (beginning, middle, end) across facial expressions, gaze, body language, and emotion, ending with a one-sentence arc summary; MELD prompts additionally instruct the model to focus on the active speaker. Inference used batch size 4 (>7 h) for IEMOCAP and batch size 2 (>10 h) for MELD. Descriptions are encoded into 768-d `[CLS]` embeddings via **BERT** [6] (truncated to 512 tokens) or **Longformer** [3] (full sequence), evaluated as separate conditions.

### 4.4 Implementation Details

**Hardware and Environment.** All experiments are conducted on an NVIDIA GeForce RTX 3060 (12 GB VRAM) using a Conda environment with PyTorch 2.8.0+cu126.

**Table 2:** Hyperparameter settings for IEMOCAP and MELD.

| Hyperparameter | IEMOCAP | MELD |
|---|---|---|
| Epochs | 120 | 50 |
| Batch Size | 16 | 16 |
| Learning Rate | $1 \times 10^{-4}$ | $7 \times 10^{-5}$ |
| Hidden Dimension | 512 | 384 |
| Sliding Window | 17 | 3 |
| Shift Window | 19 | 3 |
| Heterogeneous Graph Layers | 7 | 4 |
| Dropout Rate | 0.2 | 0.2 |
| Loss Weights | 1.0, 1.0, 0.7 | 1.0, 0.5, 0.2 |

### 4.5 Integration into Graph Based MERC Framework

**Feature Substitution and Dimensionality.** The 768-dimensional visual embeddings generated via the VLM and subsequent text encoding are substituted for the original 342-dimensional handcrafted visual descriptors in the dataset manifest. A linear projection layer is implemented at the input of the visual processing stream to map the 768-dimensional vectors into the hidden dimension space of the graph architecture. Textual and acoustic features remain identical to the GraphSmile baseline [19], with dimensions of 1024 (RoBERTa) and 1582 (openSMILE IS10) for IEMOCAP, and 1024 and 300 respectively for MELD.

**Evaluation Metrics.** Accuracy (ACC) and Weighted F1-score (WF1) serve as the primary evaluation metrics for overall model performance. The F1-score is further utilized to evaluate classification performance for each individual emotion category.

**Training and Optimization Strategy.** The model follows the multi-task optimization strategy of GraphSmile [19], jointly learning the primary emotion recognition task alongside sentiment and sentiment-shift classification. All structural components of the Graph Structure Fusion (GSF) module and the Sentiment Dynamics Perception (SDP) module are maintained as per the original implementation to ensure a controlled comparison of visual feature effectiveness.

## 5 Results and Observations

### 5.1 Results Comparison

Tables 3 and 4 report the performance of GraphSmile integrated with semantic visual representations on IEMOCAP and MELD, including overall accuracy (ACC), weighted F1 (WF1), and per-class F1 scores.

On IEMOCAP, GraphSmile with BERT-encoded VLM descriptions achieved the highest WF1 of 71.65, slightly above the baseline (71.29). Longformer encoding was comparable (WF1 71.49). BERT-based VLM features particularly improved recognition for Happy (61.49) and Frustrated (67.51) emotions.

On MELD, gains were smaller. BLIP-2 with BERT yielded the best VLM performance (ACC 67.47, WF1 66.35), while Qwen-VL-3B with BERT slightly outperformed Longformer (ACC 67.32 vs. 67.13; WF1 66.12 vs. 65.97). Overall results were close to the baseline (ACC 67.66, WF1 66.67), with minor variations across emotion categories.

**Table 3:** Main Results for the MERC Task on the IEMOCAP Dataset.

| Method | VLM Model | Text Encoder | IEMOCAP-6 | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Happy | Sad | Neutral | Angry | Excited | Frustrated | ACC | WF1 |
| COGMEN† | – | – | 51.90 | 81.70 | 68.60 | 66.00 | 75.30 | 58.20 | 68.20 | 67.60 |
| Joyful† | – | – | 60.94 | 84.42 | 68.24 | 69.95 | 73.54 | 67.55 | 70.55 | 71.03 |
| DER-GCN† | – | – | 58.80 | 79.80 | 61.50 | 72.10 | 73.30 | 67.80 | 69.70 | 69.40 |
| DialogueCRN | – | – | 53.85 | 82.66 | 71.03 | 62.33 | 77.64 | 58.81 | 68.70 | 68.82 |
| MMGCN | – | – | 47.10 | 81.91 | 66.44 | 63.51 | 76.17 | 59.06 | 67.10 | 66.81 |
| MM-DFN | – | – | 43.36 | 83.23 | 70.03 | 70.19 | 73.11 | 64.01 | 69.44 | 68.83 |
| SACL-LSTM | – | – | 51.30 | 82.25 | 71.39 | 67.78 | 75.26 | 66.94 | 70.55 | 70.60 |
| M3Net | – | – | 60.93 | 78.84 | 70.14 | 68.06 | 77.11 | 67.42 | 70.92 | 71.07 |
| GraphSmile‡ | – | – | 59.31 | 82.47 | 69.68 | 67.53 | 78.76 | 66.06 | 71.23 | 71.29 |
| GraphSmile+VLM‡ | Qwen-VL-3B | Longformer | 54.11 | 82.82 | 70.23 | 70.72 | 79.81 | 65.87 | 71.53 | 71.49 |
| GraphSmile+VLM‡ | Qwen-VL-3B | BERT | 61.49 | 83.43 | 69.69 | 67.33 | 77.12 | 67.51 | 71.53 | 71.65 |

**Table 4:** Main Results for the MERC Task on the MELD Dataset.

| Method | VLM Model | Text Encoder | MELD | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Neutral | Surprise | Fear | Sadness | Joy | Disgust | Anger | ACC | WF1 |
| Joyful† | – | – | 76.80 | 51.91 | – | 41.78 | 56.89 | – | 50.71 | 62.53 | 61.77 |
| DER-GCN† | – | – | 80.60 | 51.00 | 10.40 | 41.50 | 64.30 | 10.30 | 57.40 | 66.80 | 66.10 |
| DialogueCRN | – | – | 76.15 | 56.72 | 18.67 | 38.29 | 63.21 | 27.69 | 50.67 | 62.38 | 63.32 |
| MMGCN | – | – | 78.62 | 57.78 | 3.77 | 40.35 | 63.60 | 12.20 | 53.68 | 66.02 | 64.55 |
| MM-DFN | – | – | 79.84 | 58.43 | 15.79 | 31.65 | 64.01 | 28.04 | 53.60 | 67.05 | 65.21 |
| SACL-LSTM | – | – | 77.42 | 58.50 | 20.41 | 39.58 | 62.76 | 34.71 | 52.08 | 64.52 | 64.55 |
| M3Net | – | – | 79.14 | 59.54 | 13.33 | 42.86 | 65.05 | 21.69 | 53.54 | 66.59 | 65.83 |
| GraphSmile‡ | – | – | 80.32 | 58.96 | 18.18 | 42.46 | 64.99 | 32.43 | 53.59 | 67.66 | 66.67 |
| GraphSmile+VLM‡ | Qwen-VL-3B | Longformer | 79.82 | 57.56 | 16.44 | 41.95 | 63.94 | 31.78 | 53.19 | 67.13 | 65.97 |
| GraphSmile+VLM‡ | Qwen-VL-3B | BERT | 79.82 | 57.73 | 19.18 | 42.17 | 63.94 | 32.08 | 53.61 | 67.32 | 66.12 |
| GraphSmile+VLM‡ | BLIP2-2.7B | BERT | 79.78 | 58.30 | 14.29 | 44.20 | 64.56 | 30.77 | 54.00 | 67.47 | 66.35 |

The marker † indicates the results (%) as replicated by the GraphSmile paper, and the marker ‡ indicates the results (%) replicated by us and no marker indicates results taken from original papers.

## 5.2 Observations

The results show that semantic visual representations can effectively replace traditional handcrafted visual descriptors in graph-based MERC frameworks without significant performance loss. Modest improvements were observed, particularly on IEMOCAP where the target speaker could be clearly isolated.

The choice of text encoder affects performance depending on the length of VLM-generated descriptions. While Longformer handles sequences up to 4096

tokens, most descriptions were within BERT's 512-token limit. Consequently, BERT encoded the majority of samples efficiently, yielding slightly better performance than Longformer.

Dataset characteristics also impact the effectiveness of semantic visual features. IEMOCAP clips have relatively low spatial resolution ($360 \times 480$ after speaker-focused splitting) and short durations, limiting fine-grained facial expression visibility and temporal context for emotion inference. VLM outputs sometimes contain hallucinated or repetitive descriptions, introducing additional noise into the semantic embeddings.

On MELD, multi-party interactions complicate speaker-specific emotion recognition. Multiple characters appearing simultaneously make it difficult for the VLM to isolate the target speaker, often producing descriptions focused on the overall scene rather than the intended individual. Hallucinations and inconsistent outputs, occasionally worsened by larger batch sizes, add further noise.

Frame-level analysis with BLIP-2 [20], sampling three frames per clip (beginning, middle, end), produced more focused descriptions centered on the speaker's emotional state. This approach reduced redundancy and slightly improved performance compared to full-scene VLM descriptions, suggesting targeted reasoning on key frames can be beneficial, especially in multi-party settings.

Overall, these observations confirm that semantic visual representations are a viable alternative to handcrafted features in MERC. Their effectiveness depends on scene structure, speaker isolation, and VLM output quality. Future improvements could include explicit speaker localization, noise reduction in VLM descriptions, and optimized frame sampling for temporal emotion tracking.

## 6 Conclusion

This work explores an alternative approach to representing visual information in graph based Multimodal Emotion Recognition in Conversations (MERC). Instead of relying on conventional pixel-level visual embeddings, we employ a Vision–Language Model (VLM) to transform utterance-level video segments into structured semantic descriptions that capture observable behavioral cues and their temporal progression.

These semantically grounded representations provide a different perspective on incorporating visual signals within conversational reasoning frameworks and demonstrate that visual information can be expressed in a form closer to how emotions are typically interpreted. Our findings suggest that such semantic visual representations can serve as a viable alternative to traditional visual feature pipelines used in graph based MERC systems.

Future work may extend this idea to the acoustic modality by transforming speech signals into descriptive semantic representations that capture emotionally relevant characteristics of the audio. Additionally, architectures specifically designed to process semantic multimodal representations could further improve how such information is incorporated into conversational emotion recognition systems.

# References


1. Bai, S., Chen, K., Liu, X., Wang, J., Ge, W., Song, S., Dang, K., Wang, P., Wang, S., Tang, J., Zhong, H., Zhu, Y., Yang, M., Li, Z., Wan, J., Wang, P., Ding, W., Fu, Z., Xu, Y., Ye, J., Zhang, X., Xie, T., Cheng, Z., Zhang, H., Yang, Z., Xu, H., Lin, J.: Qwen2.5-vl technical report. arXiv preprint **abs/2502.13923** (2025). https://doi.org/10.48550/arXiv.2502.13923
2. Baltrušaitis, T., Zadeh, A., Lim, Y.C., Morency, L.P.: Openface 2.0: Facial behavior analysis toolkit. In: 2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018). pp. 59–66 (2018). https://doi.org/10.1109/FG.2018.00019
3. Beltagy, I., Peters, M.E., Cohan, A.: Longformer: The long-document transformer. In: arXiv preprint arXiv:2004.05150 (2020), https://arxiv.org/abs/2004.05150
4. Busso, C., Bulut, M., Lee, C.C., Kazemzadeh, A., Mower, E., Kim, S., Chang, J.N., Lee, S., Narayanan, S.S.: IEMOCAP: Interactive emotional dyadic motion capture database. Language Resources and Evaluation **42**(4), 335–359 (2008). https://doi.org/10.1007/s10579-008-9076-6
5. Calvo, R.A., D'Mello, S.K.: Affect detection: An interdisciplinary review of models, methods, and their applications. IEEE Transactions on Affective Computing **1**(1), 18–37 (2010). https://doi.org/10.1109/T-AFFC.2010.1
6. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL). pp. 4171–4186. Association for Computational Linguistics (2019). https://doi.org/10.18653/v1/N19-1423
7. Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: International Conference on Learning Representations (ICLR) (2021)
8. Eyben, F., Wöllmer, M., Schuller, B.: opensmile – the munich versatile and fast open-source audio feature extractor. In: Proceedings of the 18th ACM International Conference on Multimedia. pp. 1459–1462 (2010). https://doi.org/10.1145/1873951.1874246
9. Fu, Y., Wu, J., et al.: Bemerc: Behavior-aware mllm-based framework for multimodal emotion recognition in conversation. arXiv preprint arXiv:2503.23990 (2025)
10. Ghosal, D., Majumder, N., Poria, S., Chhaya, N., Gelbukh, A.: Dialoguegcn: A graph convolutional neural network for emotion recognition in conversation. In: Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). pp. 154–164. Association for Computational Linguistics (2019). https://doi.org/10.18653/v1/D19-1015
11. Hazarika, D., Poria, S., Mihalcea, R., Cambria, E., Zimmermann, R.: Icon: Interactive conversational memory network for multimodal emotion detection. In: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 2594–2604. Association for Computational Linguistics (2018). https://doi.org/10.18653/v1/D18-1280
12. Hazarika, D., Zimmermann, R., Poria, S.: MISA: Modality-invariant and -specific representations for multimodal sentiment analysis. In: Proceedings of the 28th ACM International Conference on Multimedia. pp. 1122–1131. ACM (2020). https://doi.org/10.1145/3394171.3413678

13. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 770–778 (2016). https://doi.org/10.1109/CVPR.2016.90
14. Hu, J., Liu, Y., Zhao, J., Jin, Q.: Mmgcn: Multimodal fusion via deep graph convolution network for emotion recognition in conversation. In: Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 5666–5675. Association for Computational Linguistics (2021). https://doi.org/10.18653/v1/2021.acl-long.440
15. Huang, D., Li, Q., Yan, C., Cheng, Z., Han, Z., Huang, Y., Li, X., Li, B., Wang, X., Lian, Z., Cheng, Z.Q., Peng, X.: Emotion-qwen: Training hybrid experts for unified emotion and general vision-language understanding. arXiv preprint arXiv:2505.06685 (2025)
16. Huang, G., Liu, Z., van der Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2261–2269 (2017). https://doi.org/10.1109/CVPR.2017.243
17. Kipf, T.N., Welling, M.: Semi-supervised classification with graph convolutional networks. In: International Conference on Learning Representations (ICLR) (2017)
18. Li, D., Wang, Y., Funakoshi, K., Okumura, M.: Joyful: Joint modality fusion and graph contrastive learning for multimodal emotion recognition. In: Bouamor, H., Pino, J., Bali, K. (eds.) Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing. pp. 16051–16069. Association for Computational Linguistics, Singapore (Dec 2023). https://doi.org/10.18653/v1/2023.emnlp-main.996, https://aclanthology.org/2023.emnlp-main.996/
19. Li, J., Wang, X., Zeng, Z.: Tracing intricate cues in dialogue: Joint graph structure and sentiment dynamics for multimodal emotion recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence **47**(10), 8786–8803 (2025). https://doi.org/10.1109/TPAMI.2025.3581236
20. Li, J., Li, D., Savarese, S., Hoi, S.: Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models (2023), https://arxiv.org/abs/2301.12597
21. Li, Y., Zeng, J., Shan, S., Chen, X.: Occlusion aware facial expression recognition using cnn with attention mechanism. IEEE Transactions on Image Processing **28**(5), 2439–2450 (2019). https://doi.org/10.1109/TIP.2018.2886767
22. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: RoBERTa: A robustly optimized BERT pretraining approach. arXiv preprint **abs/1907.11692** (2019). https://doi.org/10.48550/arXiv.1907.11692
23. Maas, A.L., Hannun, A.Y., Ng, A.Y.: Rectifier nonlinearities improve neural network acoustic models. In: Proc. icml. vol. 30, p. 3. Citeseer (2013)
24. Majumder, N., Poria, S., Hazarika, D., Mihalcea, R., Gelbukh, A., Cambria, E.: Dialoguernn: An attentive rnn for emotion detection in conversations. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 33, pp. 6818–6825 (2019). https://doi.org/10.1609/aaai.v33i01.33016818
25. Morency, L.P., Mihalcea, R., Doshi, P.: Towards multimodal sentiment analysis: Harvesting opinions from the web. In: Proceedings of the 13th International Conference on Multimodal Interfaces (ICMI). pp. 169–176. ACM (2011). https://doi.org/10.1145/2070481.2070509
26. Pérez-Rosas, V., Mihalcea, R., Morency, L.P.: Utterance-level multimodal sentiment analysis. In: Proceedings of the 51st Annual Meeting of the Association for

Computational Linguistics (ACL). pp. 973–982. Association for Computational Linguistics (2013)
27. Poria, S., Cambria, E., Howard, N., Huang, G.B., Hussain, A.: Fusing audio, visual and textual clues for sentiment analysis from multimodal content. Neurocomputing **174**, 50–59 (2016). https://doi.org/10.1016/j.neucom.2015.01.095
28. Poria, S., Hazarika, D., Majumder, N., Mihalcea, R.: Meld: A multimodal multi-party dataset for emotion recognition in conversations. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 527–536. Association for Computational Linguistics (2019). https://doi.org/10.18653/v1/P19-1050
29. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. In: Proceedings of the 38th International Conference on Machine Learning (ICML). pp. 8748–8763. PMLR (2021)
30. Scarselli, F., Gori, M., Tsoi, A.C., Hagenbuchner, M., Monfardini, G.: The graph neural network model. IEEE transactions on neural networks **20**(1), 61–80 (2008)
31. Schuller, B., Steidl, S., Batliner, A., et al.: Recognising realistic emotions and affect in speech: State of the art and lessons learnt from the first challenge. In: Proceedings of INTERSPEECH. pp. 149–152 (2011)
32. Shou, Y., Meng, T., Ai, W., Fu, F., Yin, N., Li, K.: A comprehensive survey on multi-modal conversational emotion recognition with deep learning (2023), https://arxiv.org/abs/2312.05735
33. Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R.: Dropout: a simple way to prevent neural networks from overfitting. The journal of machine learning research **15**(1), 1929–1958 (2014)
34. Tsai, Y.H.H., Bai, S., Liang, P.P., Kolter, J.Z., Morency, L.P., Salakhutdinov, R.: Multimodal transformer for unaligned multimodal language sequences. In: Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 6558–6569. Association for Computational Linguistics (2019). https://doi.org/10.18653/v1/P19-1656
35. Zadeh, A., Chen, M., Poria, S., Cambria, E., Morency, L.P.: Tensor fusion network for multimodal sentiment analysis. In: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 1103–1114. Association for Computational Linguistics (2017). https://doi.org/10.18653/v1/D17-1115
36. Zadeh, A., Liang, P.P., Poria, S., Cambria, E., Morency, L.P.: Multimodal language analysis in the wild: Cmu-mosei dataset and interpretable dynamic fusion graph. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL). pp. 2236–2246. Association for Computational Linguistics (2018). https://doi.org/10.18653/v1/P18-1208
37. Zadeh, A., Zellers, R., Pincus, E., Morency, L.P.: MOSI: Multimodal corpus of sentiment intensity and subjectivity analysis in online opinion videos. arXiv preprint arXiv:1606.06259 (2016)
38. Zhang, Z., Wang, W., Zhu, Y., et al.: Videmo: Affective-tree reasoning for emotion-centric video foundation models (2025), https://arxiv.org/abs/2511.02712